

Taal Chawla

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SUMMARY

AI/ML Engineer with hands-on experience building and shipping production-grade NLP systems, information retrieval pipelines, and GenAI-powered products. Built an agentic RAG pipeline delivering 20x latency reduction, 0.95 RAGAS Faithfulness, and zero cloud inference cost — end-to-end from prototyping to deployment. Strong analytical mindset with experience extracting actionable insights from structured and unstructured datasets, evaluating model outputs rigorously, and iterating until systems perform consistently beyond demos. Comfortable working independently and communicating findings clearly across technical and non-technical audiences.

EDUCATION

MAIT, GGSIPU Delhi <i>Bachelor of Technology in ECE (AI/ML) CGPA: 8.32</i>	Delhi, India Aug 2023 – May 2027
Kulachi Hansraj Model School <i>CBSE, 12th 94.4%</i>	Delhi, India Apr 2022 – Jan 2023

TECHNICAL SKILLS

Languages: Python, SQL (relational databases, complex queries, large dataset analysis), JavaScript
AI & ML Frameworks: PyTorch, HuggingFace Transformers, Scikit-learn, LangGraph, LangChain, LlamaIndex, Ollama,
Production & Deployment: FastAPI, Flask, Docker, Kubernetes, GCP, React
AI Systems & Evaluation: LLMs, RAG Pipelines, Prompt Engineering, Fine-Tuning, OpenAI API, RAGAS, Pinecone, RRF (Reciprocal Rank Fusion)

PROJECTS

- NexusOps -Agentic RAG Orchestrator** | *FastAPI, LangGraph, Docker, Kubernetes, Pinecone, Ollama* Feb 2026 – March 2026
- Engineered a workflow-native, local-first RAG pipeline using all-MiniLM-L6-v2 embeddings, cutting diagnostic retrieval latency from 240ms to sub-12ms (20x) while reducing per-query cost to \$0 via locally-orchestrated Ollama — enabling mass deployment economics with zero cloud dependency.
 - Built a multi-turn agentic chatbot on LangGraph combining LLM reasoning with deterministic rule enforcement via stateful breakpoints — intercepting 100% of unauthorized write attempts through Slack-gated human-in-the-loop at zero additional latency cost.
 - Optimized retrieval to 0.95 Faithfulness (RAGAS) using Reciprocal Rank Fusion (RRF) to synthesize dense semantic vectors with sparse hashes — systematically analyzed model outputs, identified failure modes on adversarial test cases, and iterated until validated beyond dev-time demos.
- Confidence-Aware Clinical Imaging Classification** | *Python, PyTorch, HuggingFace, Scikit-learn* Dec 2025 – Jan 2026
- Implemented post-hoc uncertainty calibration via Temperature Scaling on a fine-tuned ResNet-50, reducing Expected Calibration Error (ECE) by 75% (0.124 to 0.031) — aligning model confidence distributions with empirical accuracy for mission-critical selective prediction where overconfident errors carry clinical cost.
 - Integrated Grad-CAM heatmaps as an XAI layer into the inference pipeline, producing clinician-facing MRI activation maps to surface model reasoning and validate that correct predictions attend to clinically meaningful regions.
 - Designed and ran a systematic failure mode evaluation pipeline — iteratively probing edge cases and false negatives across augmented test sets, then diagnosing and fixing model behavior until validated beyond dev-time performance.

EXPERIENCE

Research Intern <i>DRDO (SSPL) Python, Raman Spectroscopy, Signal Analysis</i>	Jun 2025 – Aug 2025
Analyzed structural variations and vibrational modes in Silicon Carbide Polytypes (3C, 4H, 6H) for high-power ECE applications, applying structured signal extraction methodology carried directly into ML evaluation design.	

CERTIFICATIONS

Model Context Protocol: Advanced Topics <i>Anthropic — Credential ID: 7scspe8c53ka</i>	March 2026
Fine Tuning a Pretrained Model <i>Hugging Face — Skills: Large Language Models, Transformers, Fine-tuning</i>	August 2025